

WhatsApp Home DSBDA15 Logistic Regression Implementatio Social_Network_Ads/Social_Netwo

localhost:8888/notebooks/DSBDA15.ipynb

Jupyter DSBDA15 Last Checkpoint: 2 minutes ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

```
[4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix

# Load dataset
data = pd.read_csv('Social_Network_Ads.csv')

print(data.head())

# Features and target
X = data[['Age', 'EstimatedSalary']]
Y = data['Purchased']

# Split data
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.25, random_state=0)

# Scaling
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Model
model = LogisticRegression()
model.fit(X_train, Y_train)

# Prediction
Y_pred = model.predict(X_test)

# Confusion matrix
cm = confusion_matrix(Y_test, Y_pred)
```

WhatsApp Home DSBDA15 Logistic Regression Implementatio Social_Network_Ads/Social_Netwo

localhost:8888/notebooks/DSBDA15.ipynb

Jupyter DSBDA15 Last Checkpoint: 3 minutes ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

```
cm = confusion_matrix(Y_test, Y_pred)
print("Confusion Matrix:\n", cm)

# Extract values
TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

print("TP:", TP)
print("TN:", TN)
print("FP:", FP)
print("FN:", FN)

# Metrics
accuracy = (TP + TN) / (TP + TN + FP + FN)
print("Accuracy:", accuracy)

error_rate = (FP + FN) / (TP + TN + FP + FN)
print("Error Rate:", error_rate)

precision = TP / (TP + FP)
print("Precision:", precision)

recall = TP / (TP + FN)
print("Recall:", recall)

# Visualization
plt.scatter(X_test[:,0], X_test[:,1], c=Y_pred)
plt.xlabel("Age")
plt.ylabel("Estimated Salary")
plt.title("Logistic Regression Prediction")
plt.show()

User ID Gender Age EstimatedSalary Purchased
0 15624510 Male 19 19000 0
1 15810944 Male 35 20000 0
2 15668575 Female 26 43000 0
```

WhatsApp

Home

DSBDAL5

Logistic Regression Implementatio

Social_Network_Ads/Social_Netwo

localhost:8888/notebooks/DSBDAL5.ipynb

Chat

jupyter

DSBDAL5 Last Checkpoint: 3 minutes ago

Trusted

File Edit View Run Kernel Settings Help

+

✂

📄

📄

▶

⏮

⏪

⏩

⏭

Code

▼

JupyterLab

Python 3 (ipykernel)

	user_id	gender	age	estimated_salary	purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

Confusion Matrix:

```
[[65  3]
 [ 8 24]]
```

TP: 24
TN: 65
FP: 3
FN: 8
Accuracy: 0.89
Error Rate: 0.11
Precision: 0.8888888888888888
Recall: 0.75

Logistic Regression Prediction